Finite-Field Schrödinger and Dirac Dynamics from a Quadratic Lorentzian Extension

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Abstract

This paper gives a finite algebraic version of the operator core behind Schrödinger and Dirac dynamics. The setting is a prime shell $\mathbb{F}_{\mathsf{p}}$ with $\mathsf{p} = 4\mathsf{t} + 1$, so the Euclidean phase element $i_{\mathsf{t}}^2 = -1$ already lives in the base field. A nonsquare $\nu \in \mathbb{F}_{\mathsf{p}}^{\times}$ defines the Lorentzian coordinate form $Q_\nu = -\nu x_0^2 + x_1^2 + x_2^2 + x_3^2$, while the quadratic extension $K = \mathbb{F}_{\mathsf{p}}[c]/(c^2 - \nu)$ supplies Frobenius conjugation and spinorial coefficients. With these data, finite wave evolution becomes exact matrix algebra. On finite configuration spaces, translations are unitary, finite Laplacians and finite Hamiltonians are self-adjoint, and admissible Cayley steps preserve the Frobenius-Hermitian form exactly. On the Lorentzian shell $\mathbb{F}_{\mathsf{p}}^4$, explicit K-valued gamma matrices satisfy the Clifford relations, the finite Dirac operator factors exactly into a spinorial Klein-Gordon operator, and a finite $1+1$ boost subgroup transports the operator by conjugation. The paper also records exact finite counts separating reversible transport from arithmetic compression: Cayley and boost maps have loss factor 1, while the shell power map $x \mapsto x^\varepsilon$ loses distinguishability by the factor $\gcd(\varepsilon, \mathsf{p} - 1)$. The result is a compact, reproducible finite-field core of the Schrödinger-Dirac formalism: preservation, factorization, and transport are all identities of finite matrices and finite maps.

Keywords: finite fields; finite Dirac operator; Cayley transform; Clifford algebra; Frobenius involution; Lorentzian quadratic form; finite-field dynamics; reversible maps; Finite Ring Continuum

1 Introduction

Finite-field models provide a controlled setting in which kinematics, wave evolution, and covariance can be studied without completions, limits, or infinite-dimensional state spaces. They also expose a useful separation that is often hidden in continuum notation: coordinates may live in one finite field, while coefficients for amplitudes, conjugation, and spinorial operators may require a finite extension. The present paper develops this separation for a prime field

$$\mathbb{F}_{\mathsf{p}}, \qquad \mathsf{p} = 4\mathsf{t} + 1,$$

its quadratic extension

$$K = \mathbb{F}_{\mathsf{p}}[c]/(c^2 - \nu),$$

where $\nu \in \mathbb{F}_{\mathsf{p}}^{\times}$ is a nonsquare, and the Frobenius involution $\bar{z} = z^{\mathsf{p}}$ on K.

The guiding claim is structural. Once the shell, the quadratic extension, and the Frobenius involution are fixed, the relevant wave identities are finite algebraic identities. Hermitian preservation, Cayley stepping, Clifford factorization, and boost transport are expressed below as equalities of finite matrices and finite maps. All spaces, sums, operators, and transports used in the construction are finite.

This finite level removes several continuum technical burdens from the construction. There are no unbounded operator domains, no convergence questions for infinite sums, no ultraviolet regularization choices, and no measure-theoretic path integrals. What remains is the algebraic operator layer: nondegenerate Frobenius-Hermitian pairings, adjoints, form-preserving propagators, Clifford anticommutation, Dirac-to-Klein-Gordon factorization, and exact transport. What is not supplied here is a probability calculus, an ordered spectral theory, a measurement theory, scattering amplitudes, or a continuum limit theorem. Those require additional structure beyond the finite identities proved below.

The algebraic ingredients used below have standard sources. Finite fields and their quadratic extensions supply the base arithmetic [1]. Quadratic forms over finite fields, their square-class invariants, and the associated finite orthogonal and unitary groups are classical parts of finite geometry and the theory of classical groups [2, 3]. Clifford algebras and spinorial representations are algebraic constructions attached to quadratic forms, with treatments that place the gamma-matrix formalism inside the broader theory of Clifford modules and classical groups [4, 5]. Cayley-type rational transforms, including the Crank–Nicolson form of time stepping, are a standard route from self-adjoint finite-difference operators to form-preserving discrete updates [6].

Finite quantum models form a second comparison layer. Schwinger's unitary operator bases and the finite phase-space constructions of Wootters and of Gibbons, Hoffman, and Wootters show how finite algebra can carry quantum kinematics [7–9].

Galois-field and modal quantum theories study state spaces, observables, spin models, and the finite-field status of probability and spectra [10–13]. Lattice and automaton approaches to relativistic wave equations show how Weyl and Dirac dynamics can arise from exact local unitary update rules on discrete carriers [14–18].

The present paper differs from those comparison layers in scope. It does not construct a full finite-field quantum mechanics with observables and probabilities, and it does not construct a complex Hilbert-space lattice dynamics with a continuum limit. Instead, it keeps the coefficient field finite and asks which Schrödinger-Dirac operator identities survive as exact algebra over $K/\mathbb{F}_{\mathsf{p}}$. The answer is that the preservation, factorization, and transport identities can be written inside one finite coordinate/coefficient scheme.

The present construction uses this prior algebraic and finite-quantum terrain inside the FRC shell setting. In the FRC programme, a symmetry-complete prime shell is a framed finite field $\mathbb{F}_{\mathsf{p}}$ with $\mathsf{p} = 4\mathsf{t} + 1$: the additive shell, the multiplicative phase cycle, and the internal quarter-turn $i_{\mathsf{t}}^2 = -1$ coexist in the same finite arithmetic domain. The broad physical interpretation of these shells belongs to the FRC references; the present paper uses only the finite algebraic data. In that setting, symmetry-complete shells with $\mathsf{p} = 4\mathsf{t}+1$ carry an internal Euclidean phase element and a framed shell geometry [19, 20], while the Euclidean-Lorentzian split is tied to the square class of a nonsquare coefficient and its quadratic extension [21]. The condition $\mathsf{p} = 4\mathsf{t} + 1$ gives an internal Euclidean phase datum: $\mathbb{F}_{\mathsf{p}}^{\times}$ is cyclic of order $4\mathsf{t}$, and hence $\mathbb{F}_{\mathsf{p}}$ contains a square root i_{t} of -1. The nonsquare coefficient ν supplies the coordinate quadratic form

$$Q_\nu(x_0, x_1, x_2, x_3) = -\nu x_0^2 + x_1^2 + x_2^2 + x_3^2 \tag{1}$$

on $\mathbb{F}_{\mathsf{p}}^4$. The square root c of ν is not an element of $\mathbb{F}_{\mathsf{p}}$; it belongs to the quadratic extension K. This is the algebraic mechanism behind the Euclidean-Lorentzian separation established in [21]. In the present paper, the same extension is used as a coefficient field for Hermitian forms, Cayley dynamics, gamma matrices, and spinor amplitudes. The construction has two operator layers.

First, for a finite Euclidean configuration space $X = \mathbb{F}_{\mathsf{p}}^d$, the function space K^X carries a Hermitian form

$$\langle \psi, \phi \rangle_X = \sum_{x \in X} \overline{\psi(x)} \phi(x).$$

Translations are unitary, the finite Laplacian associated with a chosen ordered frame is self-adjoint,

and finite Hamiltonians of the form

$$H_{E,V,\kappa} = -\kappa\Delta_E + M_V$$

are self-adjoint. For each trace-zero parameter $\alpha \in K$ with $\bar{\alpha} = -\alpha$, the Cayley operator

$$U_{\alpha,H} = (I - \alpha H)^{-1}(I + \alpha H)$$

is defined whenever $I - \alpha H$ is invertible. It preserves the Hermitian form exactly.

Second, the coordinate shell

$$Y = \mathbb{F}_{\mathsf{p}}^4$$

with the form (1) supports a finite Dirac-type difference operator with K-valued spinors. The gamma matrices are explicit 4×4 matrices over K and satisfy

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta_{\mu\nu}I_4, \qquad \eta = \mathrm{diag}(-\nu, 1, 1, 1).$$

The corresponding Dirac operator factors exactly through a spinorial Klein-Gordon operator. A finite $1 + 1$ boost subgroup acts by exact transported Dirac operators, with an explicit spin lift.

The construction below keeps the coordinate/coefficient distinction explicit throughout. Its scope is the algebraic operator layer: Frobenius-Hermitian Schrödinger-type propagation, Clifford factorization, and Dirac-type Lorentzian transport inside one finite scheme.

The final layer records exact finite counts. For a finite map $f : X \to Y$, the image count $|f(X)|$ and loss factor $|X|/|f(X)|$ separate reversible transport from many-to-one compression. Cayley propagators and boost transports are bijections on their finite state spaces. Shell power maps $x \mapsto x^\varepsilon$ on $\mathbb{F}_{\mathsf{p}}^\times$ can be many-to-one, with exact loss factor $\gcd(\varepsilon, \mathsf{p} - 1)$. These counts are used only as finite diagnostics of reversibility. Packet transfer, invariant coding, and shell-to-shell fiber profiles are treated separately in [22].

The paper is organized as follows. Section 2 fixes the shell, coefficient, and coordinate data. Section 3 develops Hermitian function spaces and finite difference operators. Section 4 proves exact preservation for Cayley dynamics. Section 5 constructs the gamma matrices, Dirac operator, boost transport, and factorization. Section 6 records the exact reversibility counts. Sections 7 and 8 summarize the scope and the next finite-field problems.

2 Shell, Coefficient, and Coordinate Data

Let

$$\mathsf{p} = 4\mathsf{t} + 1$$

be prime. Then $\mathbb{F}_{\mathsf{p}}^\times$ is cyclic of order $4\mathsf{t}$. Therefore there is a unique subgroup of order 4, and a chosen orientation fixes an element

$$i_{\mathsf{t}} \in \mathbb{F}_{\mathsf{p}}, \qquad i_{\mathsf{t}}^2 = -1.$$

This is the finite Euclidean phase datum used below. It is the same symmetry-complete shell regime used in the FRC algebra and geometry constructions [19, 20].

Remark 2.1 (Role of the congruence condition)**.** *The condition* $\mathsf{p} \equiv 1 \pmod 4$ *is the shell condition that places the Euclidean phase element* $i_{\mathsf{t}}^2 = -1$ *inside the base field* $\mathbb{F}_{\mathsf{p}}$*. For abstract Clifford algebra alone, one can often move the required square roots into a coefficient extension. What is lost when* $\mathsf{p} \equiv 3 \pmod 4$ *is not the possibility of writing related matrices, but the FRC separation used here: the Euclidean phase is no longer internal to the base shell. The present paper works in the symmetry-complete regime where the Euclidean phase lives in* $\mathbb{F}_{\mathsf{p}}$ *and the Lorentzian nonsquare coefficient is represented by the separate extension* K*.*

The elementary arithmetic maps on the shell are

$$T_a(x) = x + a, \qquad S_m(x) = mx, \qquad P_\varepsilon(x) = x^\varepsilon, \tag{2}$$

where $a \in \mathbb{F}_\mathsf{p}$, $m \in \mathbb{F}_\mathsf{p}^\times$, and $\varepsilon \geq 1$. The maps T_a and S_m are bijections. The power map P_ε on $\mathbb{F}_\mathsf{p}^\times$ is bijective if and only if $\gcd(\varepsilon, \mathsf{p} - 1) = 1$.

Fix a nonsquare $\nu \in \mathbb{F}_\mathsf{p}^\times$ and define

$$K := \mathbb{F}_\mathsf{p}[c]/(c^2 - \nu). \tag{3}$$

Since ν is a nonsquare, $X^2 - \nu$ is irreducible over $\mathbb{F}_\mathsf{p}$, so $K \cong \mathbb{F}_{\mathsf{p}^2}$. Every element of K is written uniquely as

$$z = a + bc, \qquad a, b \in \mathbb{F}_\mathsf{p}.$$

The Frobenius involution is

$$\bar{z} := z^\mathsf{p}. \tag{4}$$

It satisfies

$$\overline{a + bc} = a - bc, \qquad \bar{c} = -c, \qquad \bar{\bar{z}} = z. \tag{5}$$

The fixed field of this involution is $\mathbb{F}_\mathsf{p}$. The trace-zero part is

$$K^- := \{\alpha \in K : \bar{\alpha} = -\alpha\} = c\mathbb{F}_\mathsf{p}. \tag{6}$$

The element c is used only on the coefficient side. The coordinate quadratic form below has coefficients in $\mathbb{F}_\mathsf{p}$ because $c^2 = \nu \in \mathbb{F}_\mathsf{p}$.

Earlier FRC terminology often describes the Lorentzian stage as the quadratic extension $K \cong \mathbb{F}_{\mathsf{p}^2}$. The present paper uses the $\mathbb{F}_\mathsf{p}$-rational coordinate shell $Y = \mathbb{F}_\mathsf{p}^4$ and uses K as the coefficient field for amplitudes, Frobenius conjugation, gamma matrices, and spin lifts. Thus the construction is the $\mathbb{F}_\mathsf{p}$-coordinate operator layer inside the quadratic-extension framework, not a replacement for the full extension viewpoint.

The Lorentzian coordinate shell is

$$Y := \mathbb{F}_\mathsf{p}^4. \tag{7}$$

It is equipped with the diagonal quadratic form

$$Q_\nu(x_0, x_1, x_2, x_3) = -\nu x_0^2 + x_1^2 + x_2^2 + x_3^2. \tag{8}$$

Let

$$\eta = \mathrm{diag}(-\nu, 1, 1, 1), \tag{9}$$

and write $\eta_{\mu\nu}$ for the entries of this fixed diagonal matrix. The subscript notation denotes matrix entries only; no index-raising convention is used.

The finite orthogonal group of the coordinate form is

$$O(Q_\nu, \mathbb{F}_\mathsf{p}) = \{\Lambda \in \mathrm{GL}_4(\mathbb{F}_\mathsf{p}) : Q_\nu(\Lambda y) = Q_\nu(y) \text{ for all } y \in Y\}.$$

The explicit finite $1 + 1$ boost subgroup used in Section 5 lies in this group.

Remark 2.2 (Coordinate and coefficient roles)**.** *The coordinate shell is $\mathbb{F}_\mathsf{p}^4$. The coefficient field for amplitudes, Hermitian conjugation, gamma matrices, and spin lifts is K. This separation is useful because the Lorentzian coefficient ν is already present in $\mathbb{F}_\mathsf{p}$, while its square root c is available only in K.*

3 Finite Hermitian Function Spaces

Let X be a finite set. Define

$$\mathcal{H}(X) := K^X,$$

the K-vector space of all K-valued functions on X.

Definition 3.1 (Hermitian form)**.** *For $\psi, \phi \in \mathcal{H}(X)$, set*

$$\langle \psi, \phi \rangle_X := \sum_{x \in X} \overline{\psi(x)} \phi(x). \tag{10}$$

Lemma 3.2. *The form (10) is Hermitian and nondegenerate.*

Proof. Hermitian symmetry follows from

$$\overline{\langle \psi, \phi \rangle_X} = \sum_{x \in X} \psi(x) \overline{\phi(x)} = \langle \phi, \psi \rangle_X.$$

For nondegeneracy, let $\psi \neq 0$. Choose $x_0 \in X$ with $\psi(x_0) \neq 0$ and define $\phi(x_0) = 1$, $\phi(x) = 0$ for $x \neq x_0$. Then

$$\langle \psi, \phi \rangle_X = \overline{\psi(x_0)} \neq 0.$$

□

Since K has no compatible order, the word Hermitian here denotes the algebraic preservation of the involutive form (10). It is not a positivity statement.

Remark 3.3 (Algebraic scope of Hermitian language)**.** *In this paper the Hermitian form supplies three pieces of finite linear algebra: a nondegenerate pairing, an adjoint operation, and the form-preservation condition $U^\dagger U = I$. Self-adjointness means $H^\dagger = H$ with respect to this pairing, and it is used to prove exact Cayley preservation in Section 4. Probabilistic interpretation, ordered expectation values, and spectral observables require additional finite-field quantum structure beyond this pairing.*

Definition 3.4 (Adjoint)**.** *Let $A : \mathcal{H}(X) \to \mathcal{H}(X)$ be K-linear. The adjoint $A^\dagger$ is the unique K-linear operator satisfying*

$$\langle A\psi, \phi \rangle_X = \langle \psi, A^\dagger \phi \rangle_X$$

for all $\psi, \phi \in \mathcal{H}(X)$.

The adjoint exists and is unique because (10) is nondegenerate and $\mathcal{H}(X)$ is finite-dimensional over K.

For $d \geq 1$, let

$$X_d := \mathbb{F}_\mathsf{p}^d$$

with its additive group law. For $u \in X_d$, define

$$(T_u \psi)(x) := \psi(x + u). \tag{11}$$

Proposition 3.5. *For every $u \in X_d$, the translation operator T_u is unitary on $\mathcal{H}(X_d)$ and*

$$T_u^\dagger = T_{-u}.$$

Proof. The map $x \mapsto x + u$ is a bijection of X_d. Reindexing gives

$$\langle T_u \psi, T_u \phi \rangle_{X_d} = \sum_{x \in X_d} \overline{\psi(x+u)} \phi(x+u) = \sum_{y \in X_d} \overline{\psi(y)} \phi(y) = \langle \psi, \phi \rangle_{X_d}.$$

Similarly,

$$\langle T_u \psi, \phi \rangle_{X_d} = \langle \psi, T_{-u} \phi \rangle_{X_d}.$$

□

Let $E = (e_1, \dots, e_d)$ be an ordered basis of X_d. Define

$$\delta_\mu^{E,+} := T_{e_\mu} - I, \tag{12}$$

$$\delta_\mu^{E,-} := T_{-e_\mu} - I, \tag{13}$$

$$\Delta_E := \sum_{\mu=1}^{d} \left(T_{e_\mu} + T_{-e_\mu} - 2I\right). \tag{14}$$

Proposition 3.6. *The finite Laplacian Δ_E is self-adjoint on $\mathcal{H}(X_d)$.*

Proof. For each μ, $T_{e_\mu}^\dagger = T_{-e_\mu}$ and $T_{-e_\mu}^\dagger = T_{e_\mu}$. Thus each summand in (14) is self-adjoint. □

Let $V : X_d \to \mathbb{F}_\mathsf{p}$. Multiplication by V defines

$$(M_V \psi)(x) := V(x)\psi(x). \tag{15}$$

Since $V(x) \in \mathbb{F}_\mathsf{p}$ is fixed by Frobenius, M_V is self-adjoint.

Definition 3.7 (Finite Hamiltonian)**.** *For $\kappa \in \mathbb{F}_\mathsf{p}$, define*

$$H_{E,V,\kappa} := -\kappa \Delta_E + M_V. \tag{16}$$

Corollary 3.8. *The Hamiltonian $H_{E,V,\kappa}$ is self-adjoint.*

Proof. Both Δ_E and M_V are self-adjoint, and $\kappa \in \mathbb{F}_\mathsf{p}$ is fixed by Frobenius. □

Definition 3.9 (Finite unitary group)**.** *For a finite set X, write*

$$U(\mathcal{H}(X)) := \{U \in \mathrm{GL}(\mathcal{H}(X)) : U^\dagger U = I\}.$$

If $|X| = n$, a choice of ordering of X identifies $\mathcal{H}(X)$ with K^n and (10) with the standard Frobenius-Hermitian form. The group $U(\mathcal{H}(X))$ is therefore the finite unitary group attached to the quadratic extension $K/\mathbb{F}_\mathsf{p}$, one of the classical finite groups of Lie type [3].

Lemma 3.10. *Let $A \in \mathrm{GL}_d(\mathbb{F}_\mathsf{p})$, let $E = (e_1, \dots, e_d)$, and let $E' = (Ae_1, \dots, Ae_d)$. Define*

$$(A_* \psi)(x) := \psi(A^{-1}x).$$

Then, for each μ,

$$T_{Ae_\mu} A_* = A_* T_{e_\mu}, \qquad \delta_\mu^{E',\pm} A_* = A_* \delta_\mu^{E,\pm}, \qquad \Delta_{E'} A_* = A_* \Delta_E.$$

Proof. For translations,

$$(T_{Ae_\mu} A_* \psi)(x) = (A_* \psi)(x + Ae_\mu) = \psi(A^{-1}x + e_\mu) = (A_* T_{e_\mu} \psi)(x).$$

The finite difference identities follow from the definitions. □

4 Cayley Dynamics on Euclidean Shells

Let $H = H_{E,V,\kappa}$ be a self-adjoint finite Hamiltonian on $\mathcal{H}(X_d)$.

Definition 4.1 (Cayley step)**.** *Let $\alpha \in K^-$, so $\bar{\alpha} = -\alpha$. If $I - \alpha H$ is invertible on $\mathcal{H}(X_d)$, define*

$$U_{\alpha,H} := (I - \alpha H)^{-1}(I + \alpha H). \tag{17}$$

Equivalently,

$$(I - \alpha H)\psi_{n+1} = (I + \alpha H)\psi_n. \tag{18}$$

Definition 4.2 (Admissible Cayley parameters)**.** *For self-adjoint H, set*

$$\mathcal{A}(H) := \{\alpha \in K^- : I - \alpha H \text{ is invertible on } \mathcal{H}(X_d)\}.$$

The set $\mathcal{A}(H)$ is finite and nonempty because $0 \in \mathcal{A}(H)$.

Remark 4.3 (Admissibility)**.** *Admissibility is governed by the finite determinant condition*

$$\det(I - \alpha H) \neq 0, \qquad \alpha \in K^-.$$

Thus the excluded trace-zero parameters are exactly the roots, inside K^-, of the finite polynomial $\det(I-\alpha H)$. The example below records one nonzero admissible parameter; a general classification of $\mathcal{A}(H)$ depends on the finite Hamiltonian.

The parameter $\alpha = 0$ always gives the identity propagator. The substantive admissibility question is therefore the nonzero part of $\mathcal{A}(H)$.

Theorem 4.4 (Exact Hermitian preservation)**.** *Let H be self-adjoint and let $\alpha \in \mathcal{A}(H)$. Then*

$$\langle U_{\alpha,H}\psi, U_{\alpha,H}\phi\rangle_{X_d} = \langle \psi, \phi\rangle_{X_d} \tag{19}$$

for all $\psi, \phi \in \mathcal{H}(X_d)$.

Proof. Since $H^\dagger = H$ and $\bar{\alpha} = -\alpha$,

$$(I - \alpha H)^\dagger = I + \alpha H, \qquad (I + \alpha H)^\dagger = I - \alpha H.$$

The invertibility of $I - \alpha H$ implies the invertibility of $I + \alpha H$ by taking adjoints. Hence

$$U_{\alpha,H}^\dagger = (I - \alpha H)(I + \alpha H)^{-1}.$$

All factors are rational functions of H, so they commute. Therefore

$$U_{\alpha,H}^\dagger U_{\alpha,H} = (I - \alpha H)(I + \alpha H)^{-1}(I - \alpha H)^{-1}(I + \alpha H) = I.$$

This gives (19). □

Corollary 4.5 (Finite periodicity)**.** *For every $\psi_0 \in \mathcal{H}(X_d)$, the Cayley orbit $\psi_n = U_{\alpha,H}^n \psi_0$ is periodic.*

Proof. The vector space $\mathcal{H}(X_d)$ is finite. The group $U(\mathcal{H}(X_d))$ is finite, and by Theorem 4.4 the operator $U_{\alpha,H}$ belongs to it. Hence some positive power of $U_{\alpha,H}$ is the identity. □

Example 4.6 (The case $\mathsf{p} = 13$)**.** *Let $\mathsf{p} = 13$, choose $\nu = 2$, and let $K = \mathbb{F}_{13}[c]/(c^2 - 2)$. Take $d = 1$, $X_1 = \mathbb{F}_{13}$, $E = (1)$, $V = 0$, and $\kappa = 1$. Then $H = -\Delta_E$. Since $K^- = c\mathbb{F}_{13}$, write $\alpha = kc$ with $k \in \mathbb{F}_{13}$. Exact computation over K gives*

$$I - kcH \text{ invertible for every } k \in \mathbb{F}_{13},$$

so $\mathcal{A}(H) = K^-$ in this example. The corresponding Cayley orders are

$$U_{0,H} = I, \qquad \operatorname{ord}(U_{kc,H}) = 13 \quad (k \in \mathbb{F}_{13}^\times).$$

Thus every vector in $\mathcal{H}(\mathbb{F}_{13})$ returns after 13 Cayley steps for each non-zero trace-zero parameter in this example. The supplementary python script [23] reproduces the full admissibility and order computation by exact arithmetic in $\mathbb{F}_{13^2}$.

The Cayley construction is an algebraic finite analogue of unitary time stepping. The preserved quantity is the Hermitian form (10); no order or probability measure is used.

5 Dirac-Type Operator and Finite Boost Transport

Let $Y = \mathbb{F}_{\mathsf{p}}^4$ with standard ordered basis $E_0 = (e_0, e_1, e_2, e_3)$ and quadratic form Q_ν from (8). A spinor field on Y is a function

$$\psi : Y \to K^4.$$

The spinor field space is

$$\mathcal{S}(Y) := (K^4)^Y.$$

For an ordered basis $E = (u_0, u_1, u_2, u_3)$ of Y, define directional forward differences

$$(\nabla_\mu^E \psi)(z) := \psi(z + u_\mu) - \psi(z), \qquad \psi \in \mathcal{S}(Y). \tag{20}$$

Because Y is an additive group, these finite differences commute for a fixed frame E.

Let

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$\sigma^2 = \begin{pmatrix} 0 & -i_{\mathsf{t}} \\ i_{\mathsf{t}} & 0 \end{pmatrix}, \qquad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Since $i_{\mathsf{t}}^2 = -1$, the Pauli identities

$$\sigma^j \sigma^k + \sigma^k \sigma^j = 2\delta_{jk} I_2$$

hold over K.

Define

$$\beta = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad \rho^k = \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix} \quad (k = 1, 2, 3),$$

and set

$$\gamma^0 := i_{\mathsf{t}} c\, \beta, \qquad \gamma^k := i_{\mathsf{t}} \rho^k \quad (k = 1, 2, 3). \tag{21}$$

Proposition 5.1 (Clifford relations)**.** *The matrices* (21) *satisfy*

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta_{\mu\nu} I_4, \qquad \eta = \mathrm{diag}(-\nu, 1, 1, 1). \tag{22}$$

Proof. The Pauli identities give $(\rho^k)^2 = -I_4$ and $\rho^j \rho^k + \rho^k \rho^j = 0$ for $j \neq k$. Also $\beta^2 = I_4$. The mixed block product is

$$\beta \rho^k = \begin{pmatrix} -\sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}, \qquad \rho^k \beta = \begin{pmatrix} \sigma^k & 0 \\ 0 & -\sigma^k \end{pmatrix},$$

so $\beta \rho^k + \rho^k \beta = 0$. Since $(i_{\mathsf{t}} c)^2 = -\nu$ and $i_{\mathsf{t}}^2 = -1$,

$$(\gamma^0)^2 = -\nu I_4, \qquad (\gamma^k)^2 = I_4,$$

and all mixed anticommutators vanish. □

5.1 Standard finite Dirac operator

Define the scalar finite Klein-Gordon operator on Y by

$$\Box_{E_0} := \sum_{\mu,\nu=0}^{3} \eta_{\mu\nu} \nabla_\mu^{E_0} \nabla_\nu^{E_0} = -\nu (\nabla_0^{E_0})^2 + \sum_{k=1}^{3} (\nabla_k^{E_0})^2. \tag{23}$$

Its spinorial extension is

$$\mathrm{KG}_{E_0}^{\mathrm{spin}} := \Box_{E_0} \otimes I_4.$$

Definition 5.2 (Standard finite Dirac operator)**.** *Define*

$$\mathcal{D}_{E_0} := \sum_{\mu=0}^{3} \gamma^\mu \nabla_\mu^{E_0} \tag{24}$$

on $\mathcal{S}(Y)$*. For* $m \in \mathbb{F}_\mathsf{p}$*, the finite Dirac equation is*

$$(\mathcal{D}_{E_0} - mI_\mathcal{S})\psi = 0. \tag{25}$$

Theorem 5.3 (Dirac-to-Klein-Gordon factorization)**.** *On* $\mathcal{S}(Y)$*,*

$$(\mathcal{D}_{E_0} - mI_\mathcal{S})(\mathcal{D}_{E_0} + mI_\mathcal{S}) = \mathrm{KG}_{E_0}^{\mathrm{spin}} - m^2 I_\mathcal{S}. \tag{26}$$

Proof. The finite differences commute, so

$$\mathcal{D}_{E_0}^2 = \sum_{\mu,\nu=0}^{3} \gamma^\mu\gamma^\nu \nabla_\mu^{E_0}\nabla_\nu^{E_0} = \frac{1}{2}\sum_{\mu,\nu=0}^{3} (\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)\nabla_\mu^{E_0}\nabla_\nu^{E_0}.$$

By Proposition 5.1, this is $\mathrm{KG}_{E_0}^{\mathrm{spin}}$. Since m is scalar, (26) follows. □

Set

$$M := \gamma^0\gamma^1.$$

Since $\gamma^0\gamma^1 = -\gamma^1\gamma^0$, $(\gamma^0)^2 = -\nu I_4$, and $(\gamma^1)^2 = I_4$,

$$M^2 = \gamma^0\gamma^1\gamma^0\gamma^1 = -\left(\gamma^0\right)^2\left(\gamma^1\right)^2 = \nu I_4.$$

Let $x, y \in \mathbb{F}_\mathsf{p}$ and assume

$$\delta := x^2 - \nu y^2 \neq 0. \tag{27}$$

Define

$$S(x,y) := xI_4 + yM. \tag{28}$$

Then

$$S(x,y)^{-1} = \delta^{-1}(xI_4 - yM). \tag{29}$$

Define

$$A := \frac{x^2 + \nu y^2}{x^2 - \nu y^2}, \qquad B := \frac{-2xy}{x^2 - \nu y^2}. \tag{30}$$

Proposition 5.4 (Spin conjugation)**.** *The spin lift* $S = S(x,y)$ *satisfies*

$$S^{-1}\gamma^0 S = A\gamma^0 + \nu B\gamma^1, \tag{31}$$

$$S^{-1}\gamma^1 S = B\gamma^0 + A\gamma^1, \tag{32}$$

$$S^{-1}\gamma^2 S = \gamma^2, \qquad S^{-1}\gamma^3 S = \gamma^3. \tag{33}$$

Proof. The inverse formula follows from $M^2 = \nu I_4$. The identities

$$M\gamma^0 = \nu\gamma^1, \qquad \gamma^0 M = -\nu\gamma^1, \qquad M\gamma^1 = \gamma^0, \qquad \gamma^1 M = -\gamma^0$$

give (31) and (32) by direct multiplication. The matrices γ^2 and γ^3 commute with $M = \gamma^0\gamma^1$, because each anticommutes with both factors γ^0 and γ^1. Hence $S^{-1}\gamma^2 S = \gamma^2$ and $S^{-1}\gamma^3 S = \gamma^3$. □

Definition 5.5 (Coordinate boost)**.** *For $x, y \in \mathbb{F}_{\mathsf{p}}$ satisfying (27), define*

$$\Lambda(x,y) := \begin{pmatrix} A & B & 0 & 0 \\ \nu B & A & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{34}$$

Proposition 5.6. *The matrix $\Lambda(x,y)$ belongs to $O(Q_\nu, \mathbb{F}_{\mathsf{p}})$.*

Proof. A direct calculation from (30) gives

$$A^2 - \nu B^2 = 1.$$

For the displayed 2×2 block, this identity is exactly $\Lambda^T \eta \Lambda = \eta$. The last two coordinates are fixed. □

Proposition 5.7 (Boost subgroup)**.** *The set*

$$G_\nu := \{\Lambda(x,y) : x, y \in \mathbb{F}_{\mathsf{p}},\ x^2 - \nu y^2 \neq 0\}$$

is a subgroup of $O(Q_\nu, \mathbb{F}_{\mathsf{p}})$.

Proof. Let $z = x + yc$ and $w = u + vc$ in $K^\times$. Their product is

$$zw = (xu + \nu yv) + (xv + yu)c.$$

The norm satisfies

$$(xu + \nu yv)^2 - \nu(xv + yu)^2 = (x^2 - \nu y^2)(u^2 - \nu v^2).$$

Direct multiplication of the matrices (34) gives

$$\Lambda(x,y)\Lambda(u,v) = \Lambda(xu + \nu yv, xv + yu).$$

Also $\Lambda(1,0) = I_4$ and $\Lambda(x,y)^{-1} = \Lambda(x,-y)$. Hence, G_ν is a subgroup. □

5.2 Transported Dirac operators

Let $\Lambda = \Lambda(x,y) \in G_\nu$ and set

$$E_\Lambda = (\Lambda e_0, \Lambda e_1, \Lambda e_2, \Lambda e_3).$$

Define the pullback on spinor fields by

$$(\Lambda_* \psi)(z) := \psi(\Lambda^{-1} z). \tag{35}$$

Lemma 5.8. *For every μ,*

$$\nabla_\mu^{E_\Lambda} \Lambda_* = \Lambda_* \nabla_\mu^{E_0}. \tag{36}$$

Proof. For $\psi \in \mathcal{S}(Y)$ and $z \in Y$,

$$(\nabla_\mu^{E_\Lambda} \Lambda_* \psi)(z) = (\Lambda_* \psi)(z + \Lambda e_\mu) - (\Lambda_* \psi)(z) = \psi(\Lambda^{-1} z + e_\mu) - \psi(\Lambda^{-1} z),$$

which is $(\Lambda_* \nabla_\mu^{E_0} \psi)(z)$. □

For $S = S(x,y)$, define the spinor transport

$$\mathcal{T}_{\Lambda,S} := S \Lambda_*. \tag{37}$$

The transported Dirac operator is

$$\mathcal{D}_{\Lambda,S} := \mathcal{T}_{\Lambda,S} \mathcal{D}_{E_0} \mathcal{T}_{\Lambda,S}^{-1}. \tag{38}$$

Proposition 5.9 (Explicit transported form)**.** *The operator* (38) *is*

$$\mathcal{D}_{\Lambda,S} = \sum_{\mu=0}^{3} \widehat{\gamma}^{\mu}_{\Lambda,S} \nabla^{E_\Lambda}_{\mu}, \tag{39}$$

where

$$\widehat{\gamma}^{0}_{\Lambda,S} = A\gamma^0 - \nu B\gamma^1, \tag{40}$$
$$\widehat{\gamma}^{1}_{\Lambda,S} = -B\gamma^0 + A\gamma^1, \tag{41}$$
$$\widehat{\gamma}^{2}_{\Lambda,S} = \gamma^2, \qquad \widehat{\gamma}^{3}_{\Lambda,S} = \gamma^3. \tag{42}$$

Proof. Since S is constant on Y, it commutes with the difference operators. Using (36),

$$\mathcal{D}_{\Lambda,S} = S\Lambda_* \left(\sum_{\mu} \gamma^\mu \nabla^{E_0}_{\mu} \right) \Lambda_*^{-1} S^{-1} = S \left(\sum_{\mu} \gamma^\mu \nabla^{E_\Lambda}_{\mu} \right) S^{-1}.$$

Thus the transported gamma matrices are $S\gamma^\mu S^{-1}$. Replacing y by $-y$ in Proposition 5.4 gives (40)–(42). □

Corollary 5.10 (Transported Clifford relations)**.** *For every $\Lambda \in G_\nu$ and spin lift S, the family $\widehat{\gamma}^{\mu}_{\Lambda,S}$ satisfies*

$$\widehat{\gamma}^{\mu}_{\Lambda,S}\widehat{\gamma}^{\nu}_{\Lambda,S} + \widehat{\gamma}^{\nu}_{\Lambda,S}\widehat{\gamma}^{\mu}_{\Lambda,S} = 2\eta_{\mu\nu} I_4.$$

Proof. Conjugation by S preserves anticommutators. □

For the boosted frame, set

$$\Box_{E_\Lambda} := \sum_{\mu,\nu=0}^{3} \eta_{\mu\nu} \nabla^{E_\Lambda}_{\mu} \nabla^{E_\Lambda}_{\nu} = -\nu(\nabla^{E_\Lambda}_{0})^2 + \sum_{k=1}^{3} (\nabla^{E_\Lambda}_{k})^2 \tag{43}$$

and

$$\mathrm{KG}^{\mathrm{spin}}_{E_\Lambda} := \Box_{E_\Lambda} \otimes I_4.$$

Theorem 5.11 (Transported factorization)**.** *For every $\Lambda \in G_\nu$ and spin lift S,*

$$(\mathcal{D}_{\Lambda,S} - mI_{\mathcal{S}})(\mathcal{D}_{\Lambda,S} + mI_{\mathcal{S}}) = \mathrm{KG}^{\mathrm{spin}}_{E_\Lambda} - m^2 I_{\mathcal{S}}. \tag{44}$$

Proof. Use Proposition 5.9, the commutativity of the finite differences in the frame E_Λ, and the transported Clifford relations. The calculation is identical to the proof of Theorem 5.3. □

Corollary 5.12 (Exact covariance of the transported family)**.** *For $\Lambda \in G_\nu$ with spin lift S and every $\psi \in \mathcal{S}(Y)$,*

$$\mathcal{D}_{\Lambda,S}(\mathcal{T}_{\Lambda,S}\psi) = \mathcal{T}_{\Lambda,S}(\mathcal{D}_{E_0}\psi). \tag{45}$$

Consequently, a solution of $(\mathcal{D}_{E_0} - mI_{\mathcal{S}})\psi = 0$ is transported to a solution of

$$(\mathcal{D}_{\Lambda,S} - mI_{\mathcal{S}})\mathcal{T}_{\Lambda,S}\psi = 0.$$

Proof. By Proposition 5.9, $\mathcal{D}_{\Lambda,S}$ is the explicit finite difference operator obtained by transporting the standard gamma family and frame. Since it is also defined by the conjugation relation (38), (45) follows immediately. The equation with mass follows because m is scalar and commutes with $\mathcal{T}_{\Lambda,S}$. □

Example 5.13 (The case $\mathsf{p} = 13$)**.** *Let* $\mathsf{p} = 13$ *and* $\nu = 2$*. Taking* $x = y = 1$ *gives*

$$A = 10, \qquad B = 2$$

in $\mathbb{F}_{13}$*, hence*

$$\Lambda(1,1) = \begin{pmatrix} 10 & 2 & 0 & 0 \\ 4 & 10 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in O(Q_2, \mathbb{F}_{13}).$$

The spin lift is $S(1,1) = I_4 + \gamma^0\gamma^1$*, and the transported gamma matrices are*

$$\widehat{\gamma}^0 = 10\gamma^0 - 4\gamma^1, \qquad \widehat{\gamma}^1 = -2\gamma^0 + 10\gamma^1,$$

with $\widehat{\gamma}^2 = \gamma^2$ *and* $\widehat{\gamma}^3 = \gamma^3$*. The supplementary python script [23] verifies the Clifford relations, conjugation formulas, and a sample covariance identity by exact arithmetic in* $\mathbb{F}_{13^2}$*.*

6 Exact Reversibility Counts

This section is diagnostic. It records exact finite cardinality facts that distinguish reversible maps from compressive maps. It does not define a thermodynamic entropy, a probability measure on amplitudes, or a packet-level transfer theory.

Definition 6.1 (Image count and loss factor)**.** *Let* $f : X \to Y$ *be a map of finite sets. Define*

$$I_X(f) := |f(X)|, \qquad L_X(f) := \frac{|X|}{|f(X)|} \in \mathbb{Q}.$$

The image count records how many states remain distinguishable after applying f. The loss factor is an exact finite ratio. It is used here only for finite maps.

Proposition 6.2. *For a self-map* $f : X \to X$ *of a finite set, the following are equivalent:*

1. f *is bijective;*
2. $I_X(f) = |X|$*;*
3. $L_X(f) = 1$*.*

Proof. If f is bijective, then $f(X) = X$. Conversely, if $|f(X)| = |X|$, then f is surjective and hence bijective because X is finite. The equivalence with $L_X(f) = 1$ follows from the definition. □

Proposition 6.3 (Power-map count)**.** *Let* $X = \mathbb{F}_{\mathsf{p}}^{\times}$ *and let*

$$P_\varepsilon(x) = x^\varepsilon, \qquad \varepsilon \geq 1.$$

Set $d = \gcd(\varepsilon, \mathsf{p} - 1)$*. Then*

$$I_X(P_\varepsilon) = \frac{\mathsf{p} - 1}{d}, \qquad L_X(P_\varepsilon) = d.$$

Each nonempty fiber has cardinality d*.*

Proof. Choose a generator g of the cyclic group $\mathbb{F}_{\mathsf{p}}^{\times}$. Every element is g^k with $k \in \mathbb{Z}/(\mathsf{p} - 1)\mathbb{Z}$, and

$$P_\varepsilon(g^k) = g^{\varepsilon k}.$$

Thus P_ε is multiplication by ε on $\mathbb{Z}/(\mathsf{p} - 1)\mathbb{Z}$. Its kernel has size d and its image has size $(\mathsf{p} - 1)/d$. □

Corollary 6.4 (Cayley propagators)**.** *For $\alpha \in \mathcal{A}(H)$, the Cayley propagator $U_{\alpha,H} : \mathcal{H}(X_d) \to \mathcal{H}(X_d)$ has loss factor* 1.

Proof. The operator $U_{\alpha,H}$ is invertible by definition. Apply Proposition 6.2 to the underlying finite set $\mathcal{H}(X_d)$. □

Corollary 6.5 (Boost transport)**.** *For $\Lambda \in G_\nu$ and spin lift S, the transport*

$$\mathcal{T}_{\Lambda,S} = S\Lambda_* : \mathcal{S}(Y) \to \mathcal{S}(Y)$$

has loss factor 1.

Proof. The pullback Λ_* is bijective because Λ is invertible on Y, and multiplication by S is bijective because S is invertible. Apply Proposition 6.2. □

Thus the wave operators constructed above are reversible finite maps. The shell power maps provide a contrasting finite source of arithmetic compression, with exact loss factor $\gcd(\varepsilon, \mathsf{p} - 1)$. The supplementary python script [23] reproduces the finite examples by exact arithmetic over $\mathbb{F}_{13}[c]/(c^2 - 2)$.

7 Discussion

The construction gives a finite operator layer over the Euclidean-Lorentzian shell data. The Euclidean side uses $\mathbb{F}_\mathsf{p}^d$ as configuration space and K as coefficient field. The Lorentzian side uses $Y = \mathbb{F}_\mathsf{p}^4$ as coordinate shell and the same coefficient field K for gamma matrices and spinors. The coordinate/coefficient split is essential: Q_ν is written over $\mathbb{F}_\mathsf{p}$, while the square root c of the nonsquare ν belongs to K.

The Schrödinger-type result is an exact Cayley preservation theorem. A self-adjoint finite Hamiltonian generates a finite family of admissible Cayley maps, and each such map preserves the Hermitian form exactly. Since the state space is finite, every Cayley trajectory is periodic. This is an algebraic periodicity statement over a finite field; it is independent of any continuum spectral theorem.

The Dirac-type result is an exact Clifford and transport construction. The gamma matrices encode the diagonal form $\eta = \mathrm{diag}(-\nu, 1, 1, 1)$, the finite Dirac operator factors to the spinorial Klein-Gordon operator, and the explicit boost subgroup G_ν transports the standard operator to conjugate finite operators. The covariance statement is therefore an equality of finite operators on the same finite spinor-field set.

The finite counts in Section 6 distinguish two algebraic behaviors. Cayley propagators and boost transports are bijections; shell power maps need not be. This is the finite-set content behind the loss factors. The packet, invariant, and fiber structure associated with shell changes is a separate finite-transfer problem developed in [22].

The finite setting also fixes the level at which the operator identities live. The Hermitian form is a finite sum, the Hamiltonians and Dirac operators are endomorphisms of finite-dimensional K-spaces, Cayley preservation is a matrix identity whenever the denominator is invertible, and the Dirac-to-Klein-Gordon factorization is an exact equality of finite difference operators. This is the algebraic core carried by the construction.

This is the main consolidation supplied by the paper. The finite scheme does not approximate the continuum formalism by adding cutoffs; it isolates the part of the formalism that is already algebraic and makes it exact inside finite arithmetic. The price of this exactness is also explicit: ordered probabilities, continuum spectra, scattering theory, and measurement interpretation are not consequences of the present identities alone.

Several mathematical problems are now well posed. The first is to classify self-adjoint Hamiltonians over $\mathbb{F}_\mathsf{p}^d$ up to frame change and compute the exact orders of their Cayley propagators.

The second is to extend the explicit $1+1$ boost subgroup to larger finite orthogonal and spinorial families. The third is to connect the Dirac transport constructed here with packet-level invariants in the quadratic extension. The fourth is to formulate an observer-local coordinate derivation of the finite Dirac arena from the global shell geometry of [20].

It should be also noted that the programme-level ontological position behind this construction is to treat the primary object as a symmetry space of finite algebraic possibilities, realized through a collection of finite fields. This resonates with Kastner's possibilist transactional interpretation, where quantum possibility is treated as physically significant rather than as a merely calculational device [24]. A local quantum system is therefore not treated as a self-standing object in the strongest sense. It is a transient, quasi-stable sparse realization of a larger finite structure. The only wave function with direct ontological scope is the universe-scale one; local wave functions are effective projections that persist only insofar as the corresponding sparse realization remains stable over a finite interval of evolution.

8 Conclusion

For primes $\mathsf{p} = 4\mathsf{t} + 1$, the finite field $\mathbb{F}_\mathsf{p}$ carries an internal Euclidean phase element $i_\mathsf{t}^2 = -1$. A nonsquare $\nu \in \mathbb{F}_\mathsf{p}^\times$ determines the Lorentzian coordinate form Q_ν on $\mathbb{F}_\mathsf{p}^4$, while the quadratic extension $K = \mathbb{F}_\mathsf{p}[c]/(c^2 - \nu)$ supplies Frobenius conjugation and spinorial coefficients. From these data one obtains exact finite analogues of Schrödinger and Dirac dynamics.

On Euclidean configuration spaces $\mathbb{F}_\mathsf{p}^d$, finite Hamiltonians built from translations and finite potentials are self-adjoint, and admissible Cayley transforms preserve the Hermitian form exactly. On the Lorentzian coordinate shell $\mathbb{F}_\mathsf{p}^4$, explicit gamma matrices over K satisfy the Clifford relations for $\eta = \mathrm{diag}(-\nu, 1, 1, 1)$, and the finite Dirac difference operator factors into a spinorial Klein-Gordon operator. A finite $1+1$ boost subgroup admits an explicit spin lift and transports the Dirac operator by exact conjugation.

The resulting maps are finite and reproducible. Cayley propagators and boost transports are bijective on their finite state spaces. By contrast, shell power maps on $\mathbb{F}_\mathsf{p}^\times$ compress by the exact arithmetic factor $\gcd(\varepsilon, \mathsf{p} - 1)$. This gives a compact finite-field distinction between reversible transport and arithmetic compression.

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